Methodology: Hierarchical Mycosis Fungoides Classification System

1. Overview

A hierarchical deep learning framework was developed to diagnose Mycosis Fungoides (MF) and differentiate it from disease mimics using multi-magnification histopathological image analysis combined with clinical features. The system employs **late fusion** of two independent convolutional neural networks trained at different microscopic magnifications (10× and 20×), integrated with a machine learning-based clinical notes classifier.

> **Magnification Terminology:** Throughout this document, "10×" and "20×" refer to the power of the **objective lens** (near the slide). The microscope uses a fixed 10× eyepiece lens (near the camera/observer), so these correspond to **total optical magnifications of 100× and 200×**, respectively.

2. Histopathological Image Analysis

2.1 Preprocessing Pipeline

2.1.1 Patch Extraction

Whole slide images were preprocessed using a sliding window approach to extract diagnostically relevant tissue patches:

10× Magnification:

- Patch size: 512 × 512 pixels
- Stride: 256 pixels (50% overlap)
- Maximum patches per image: 150

20× Magnification:

- Patch size: 1024 × 1024 pixels

- Stride: 512 pixels (50% overlap)
- Maximum patches per image: 150

2.1.2 Tissue Foreground Detection

To eliminate non-diagnostic regions (background, artifacts, staining anomalies), each patch underwent automated quality filtering:

Process:

1. Convert patch from RGB to HSV color space
2. Extract saturation channel to identify stained tissue
3. Apply threshold: foreground pixels = saturation > 20
4. Compute foreground ratio:

$$\text{fg\_ratio} = \frac{N_{\text{foreground}}}{N_{\text{total}}}$$

5. **Inclusion criterion**: $\text{fg\_ratio} \geq 0.285$ (28.5% minimum tissue threshold)

This tissue detection mechanism ensures only patches containing sufficient cellular material are processed, automatically rejecting background and edge artifacts.

2.1.3 Normalization

All patches were normalized using ImageNet pretraining statistics:

$$\text{Normalized Patch} = \frac{\text{Original} - \mu}{\sigma}$$

Where:

- **Mean**: $\mu = [0.485, 0.456, 0.406]$
- **Std Dev**: $\sigma = [0.229, 0.224, 0.225]$

2.2 Deep Learning Architecture: Dual-Magnification Ensemble

A two-branch architecture was implemented with independent models trained for each magnification level, capturing complementary diagnostic information.

2.2.1 10× Classifier (Architectural Features)

Parameter	Value
Base Architecture	TF-EfficientNet-B3
Input Resolution	512 × 512 pixels
Output Classes	5 (MF, PLEVA-PLC, Pseudo-Lymphoma, T-cell dyscrasia, B-cell Lymphoma)
Dropout Rate	0.40
Inference Batch Size	8 patches
Model Parameters	~10.3M

Rationale: The 10× magnification captures broader tissue architecture and glandular distribution patterns, enabling assessment of infiltration depth and architectural distortion characteristic of MF.

Training Configuration (shared with 20× model):

- Loss function: Dynamically weighted Cross-Entropy (class weights inversely proportional to training distribution)
- Optimizer: AdamW (static learning rate: 2×10^{-4}, weight decay: 0.01)
- Augmentations: Random horizontal/vertical flips, random rotation (±15°), color jittering (brightness/contrast/saturation ±20%, hue ±5%)
- Mixed precision: Automatic Mixed Precision (AMP)
- Gradient accumulation: 1 step
- Validation metric: F2-Score (emphasizing sensitivity)
- Normalization: ImageNet statistics

2.2.2 20× Classifier (Cytological Features)

Parameter	Value
Base Architecture	TF-EfficientNet-B3
Input Resolution	512 × 512 pixels
Native Patch Size	1024 × 1024 pixels
Output Classes	5

Parameter	Value
Dropout Rate	0.40
Inference Batch Size	2 patches
Model Parameters	~10.3M

Rationale: The 20× magnification emphasizes cytological details—nuclear morphology, chromatin patterns, mitotic figures, and immune infiltrates—essential for distinguishing MF from benign disease mimics.

Training Configuration (shared with 10× model):

- Loss function: Dynamically weighted Cross-Entropy (same as 10×)
- Optimizer: AdamW (static learning rate: 2×10^{-4}, weight decay: 0.01)
- Augmentations: Same spatial and color augmentation pipeline as 10×
- Mixed precision: Automatic Mixed Precision (AMP)
- Gradient accumulation: 8 steps (to compensate for smaller batch size)
- Validation metric: F2-Score (emphasizing sensitivity)

2.3 Patch-to-Patient Aggregation: Mean Probability

Raw patch-level outputs from both models were aggregated to patient-level predictions using **mean probability averaging**:

$$P_{\text{patient}}(c) = \frac{1}{N}\sum_{i=1}^{N} P_{\text{patch}_i}(c)$$

Where:

- $P_{\text{patient}}(c)$ = patient-level probability for class c
- N = total number of extracted patches
- $P_{\text{patch}_i}(c)$ = softmax output for patch i and class c

Justification: This approach assumes that aggregate patch characteristics reflect overall patient pathology, providing robustness against local artifacts while preserving diagnostic signal across the tissue sample.

Output: While the individual models can output 5-class probability vectors, for the purpose of the binary fusion (MF vs Non-MF), we extract the scalar **Non-MF probability** for each patient:

- $P^{10x}_{\text{non-MF}} \in [0, 1]$
- $P^{20x}_{\text{non-MF}} \in [0, 1]$

3. Decision Thresholding Strategy: Youden's J Optimization

Rather than defaulting to 0.5, optimal decision thresholds were computed independently for each model using **Youden's J statistic**, which balances sensitivity and specificity:

$$J(\theta) = \text{TPR}(\theta) - \text{FPR}(\theta) = \text{Sensitivity} + \text{Specificity} - 1$$

The optimal threshold maximizes this statistic:

$$\theta^* = \arg\max_{\theta} J(\theta)$$

3.1 10× Model Threshold

- **Optimal Threshold**: $\theta^*_{10x} = 0.50$
- **Derivation**: The 10× model was naturally balanced at a 0.5 decision boundary
- **Decision Rule**: $P(\text{Non-MF}) > 0.50 \rightarrow \text{Non-MF}$; otherwise $\rightarrow \text{MF}$
- **Test Set Performance**: Accuracy: 80.60% | Sensitivity: 80.43% | Specificity: 80.43%

3.2 20× Model Threshold

- **Optimal Threshold**: $\theta^*_{20x} = 0.8351$
- **Derivation**: Youden's J maximization on validation cohort
- **Decision Rule**: $P(\text{Non-MF}) > 0.8351 \rightarrow \text{Non-MF}$; otherwise $\rightarrow \text{MF}$
- **At default 0.5 threshold**: MF sensitivity was an unacceptable **43.48%** (severe conservative bias)
- **Test Set Performance at 0.8351**: Accuracy: 83.58% | Sensitivity: 89.13% | Specificity: 89.13%

Clinical Interpretation: The elevated 20× threshold (0.8351 vs 0.50) reflects that cytological mimicry is more prevalent; hence stronger confidence is required to exclude MF. This asymmetry protects against false negatives—critical in cancer diagnosis.

4. Late Fusion with Probability Calibration

4.1 The Calibration Problem

Directly averaging probabilities from both models introduces systematic bias because they operate under different decision calibrations:

- 10× model: 0.5 ≈ indifference point
- 20× model: 0.8351 ≈ indifference point

Without correction, the fused probabilities would be skewed toward the 20× model's calibration.

4.2 Recalibration Transformation

The 20× Non-MF probability is recalibrated to align with the standard 0.5 scale using **piecewise-linear transformation**:

$$p_{20x}^{\text{non-MF, cal}} = \begin{cases} p_{20x}^{\text{non-MF}} \cdot \dfrac{0.5}{0.8351} & \text{if } p_{20x}^{\text{non-MF}} < 0.8351 \\ 0.5 + \left(p_{20x}^{\text{non-MF}} - 0.8351\right) \cdot \dfrac{0.5}{1.0 - 0.8351} & \text{if } p_{20x}^{\text{non-MF}} \geq 0.8351 \end{cases}$$

Transformation Effects:

1. Maps $[0, 0.8351] \rightarrow [0, 0.5]$ (stretching lower probabilities)
2. Maps $[0.8351, 1.0] \rightarrow [0.5, 1.0]$ (stretching upper probabilities)
3. Ensures $0.8351 \rightarrow 0.5$ (calibration point alignment)

The Simple Explanation & Real-World Example

Imagine you have two teachers grading the same exam.

- **Teacher A (The 10x Model)** is a standard grader. If you get a 50%, you pass.
- **Teacher B (The 20x Model)** is a notoriously harsh grader. Because their test is so skewed, their passing line is an 83.51%. If you get an 83.51%, you pass.

If you want to average a student's grades from both teachers fairly, you cannot just add their raw scores together. A 75% means "Failing" to Teacher B, but "Passing" to Teacher A. Averaging them directly would ruin the math.

To fix this, you have to recalibrate Teacher B's grades. You take their harsh passing line (83.51%) and mathematically stretch/shift it so it aligns with the standard passing line (50%). Now both teachers use

50% as the passing mark, and you can safely average their scores.

The "Accuracy Drop" happens because when you stretch and bend Teacher B's grading scale to make it fit the standard scale, a student who was right on the borderline might slightly shift sides. In your data, exactly one borderline patient out of 67 flipped from correct to incorrect because of this stretching, dropping your accuracy from 83.58% (56/67 right) to 82.09% (55/67 right).

Result: Both models now operate on comparable probability scales, enabling valid weighted fusion.

4.3 Weighted Fusion via Brier Score Optimization

The calibrated Non-MF probabilities are combined via weighted averaging:

$$P_{\text{fused}}^{\text{non-MF}} = w \cdot P_{10x}^{\text{non-MF}} + (1-w) \cdot P_{20x}^{\text{non-MF, cal}}$$

The Brier Score (mean squared prediction error) was tracked to measure calibration:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} (P_{\text{fused},i} - y_i)^2$$

Rationale for Brier Score: Unlike accuracy (binary outcome), Brier score penalizes both:

- Incorrect predictions
- Overconfident correct predictions

In medical AI, this prevents dangerous overconfidence in predictions. While the strictly optimal weight minimizing the Brier Score was found to be $w = 0.60$ (Brier: 0.1401), a **clinical golden weight** of $w^* = 0.20$ was manually selected to prioritize sensitivity.

4.3.1 Weight Sweep Results

Systematic evaluation across $w \in [0.0, 1.0]$ with step 0.05. A subset of key intervals is shown:

Weight	Brier Score	Accuracy	Sensitivity (MF)	Specificity (MF)	ROC-AUC	F2 (MF)
0.00	0.1526	82.09%	89.13%	89.13%	0.8509	0.8836
0.10	0.1486	83.58%	89.13%	89.13%	0.8634	0.8874
0.20	**0.1455**	**83.58%**	**89.13%**	**89.13%**	**0.8778**	**0.8874**

Weight	Brier Score	Accuracy	Sensitivity (MF)	Specificity (MF)	ROC-AUC	F2 (MF)
0.40	0.1413	77.61%	80.43%	80.43%	0.8882	0.8150
0.60	*0.1401*	*79.10%*	*80.43%*	*80.43%*	*0.8861*	*0.8186*
0.80	0.1418	80.60%	80.43%	80.43%	0.8789	0.8222
1.00	0.1466	80.60%	80.43%	80.43%	0.8685	0.8222

Selected Clinical Weight: $w^* = 0.20$
Interpretation:

- **20% contribution** from 10× architectural model
- **80% contribution** from 20× cytological model
- **Clinical reasoning**: While architectural patterns provide context, cytological details are more distinctive for MF diagnosis. The weight $w = 0.20$ maximizes MF Sensitivity (89.13%) and F2-Score (0.8874) without compromising overall Accuracy (83.58%), providing the safest clinical threshold.
- **Performance at w=0.20**: Brier: 0.1455 | Acc: 83.58% | Sens: 89.13% | Spec: 89.13% | AUC: 0.8778

5. Multi-Class Prediction and Binary Extraction

5.1 Native Multi-Class Prediction

The deep learning models are trained as 5-class classifiers. At inference time, the model outputs a 5-class probability vector via a softmax activation. For patch-level and individual model evaluations, the predicted class is determined directly by selecting the class with the highest probability:

$$\hat{y}_{\text{multi}} = \arg \max_{c \in \{0,1,2,3,4\}} P(c)$$

Class Mapping (5-way):

- Index 0: **B-cell Lymphoma**
- Index 1: **Mycosis Fungoides (MF)**
- Index 2: **PLEVA-PLC**
- Index 3: **T-cell dyscrasia**

- Index 4: **Pseudolymphoma**

5.2 Binary Extraction for Late Fusion

While the models natively output 5 classes, the primary clinical objective and the late-fusion pipeline focus on the binary distinction between MF and Non-MF conditions.

To enable this binary fusion:

1. The probabilities of all non-MF classes (Indices 0, 2, 3, 4) are summed (or extracted) to form a scalar $P_{\text{non-MF}}$ for each patch.
2. These patch-level binary probabilities are mean-aggregated to form the patient-level $P_{\text{patient}}^{\text{non-MF}}$.
3. The late-fusion pipeline then combines these binary probabilities across magnifications using the recalibration and weighting strategy described in Section 4.

The final binary diagnostic decision is made using the fused scalar:

$$\hat{y}_{\text{binary}} = \begin{cases} \text{MF} & \text{if } P_{\text{fused}}^{\text{non-MF}} \leq 0.5 \\ \text{Non-MF} & \text{if } P_{\text{fused}}^{\text{non-MF}} > 0.5 \end{cases}$$

6. Clinical Features Classifier: Machine Learning Module

6.1 Clinical Feature Engineering

A complementary classifier was trained on patient metadata to leverage diagnostic information beyond histology.

6.1.1 Feature Selection (16 Selected Features)

Univariate statistical analysis was performed on 22 clinical variables. Continuous variables were assessed using the Mann-Whitney U test (due to skewed distributions), and categorical variables were evaluated via Chi-Squared tests. Features with $p < 0.05$ were selected, yielding **15 statistically significant features**. One additional feature (Nodule, $p = 0.2007$) was deliberately retained for its staging value, bringing the total to **16 features**:

Feature	Type	Test Used	p-value	Domain
Macules	Categorical	Chi-Squared	< 0.001	Morphology
Biopsy 1 Morphology	Categorical	Chi-Squared	< 0.001	Biopsy findings
Papules	Categorical	Chi-Squared	< 0.001	Morphology
Age	Continuous	Mann-Whitney U	< 0.001	Demographics
Lesion Color	Categorical	Chi-Squared	< 0.001	Lesion phenotype
Patch	Categorical	Chi-Squared	< 0.001	Morphology
Biopsy 2 Morphology	Categorical	Chi-Squared	< 0.001	Biopsy findings
Visit Type	Categorical	Chi-Squared	< 0.001	Visit context
Duration (Months)	Continuous	Mann-Whitney U	< 0.001	Disease history
Plaque	Categorical	Chi-Squared	< 0.001	Morphologic stage
Scales	Categorical	Chi-Squared	< 0.001	Surface features
Site: Head/Neck	Categorical	Chi-Squared	0.0001	Anatomic distribution
Site: Lower Limbs	Categorical	Chi-Squared	0.009	Anatomic distribution
Disease Course	Categorical	Chi-Squared	0.01	Disease trajectory
Biopsy 2 Site	Categorical	Chi-Squared	0.0397	Biopsy location
Nodule†	Categorical	Chi-Squared	**0.2007**	Morphologic stage

†**Nodule Retention Rationale:** Although statistically insignificant for the binary MF vs. Non-MF diagnosis task ($p = 0.2007$), the Nodule feature was deliberately retained because it carries critical prognostic value for intra-MF staging: the presence of nodules is a near-deterministic indicator of the Tumor stage, functioning as an almost rule-based marker. Excluding it would sacrifice essential staging information for a marginal gain in diagnostic feature parsimony.

Selection Rationale: Features selected based on statistical significance ($p < 0.05$) and clinical plausibility (confirmed by dermatopathology experts).

6.1.2 Data Preprocessing

- **Continuous features**: Median imputation for missingness
- **Categorical features**: Label encoding (alphabetical order)

- **Binary features**: 0/1 encoding (No=0, Yes=1)
- **Class weighting**: $\text{scale\_pos\_weight} = 12$ (addressing MF/Non-MF imbalance)

6.2 Three-Model Comparison Framework

Three distinct algorithms were systematically evaluated using **5-fold stratified cross-validation** to identify the optimal clinical classifier:

6.2.1 Model 1: XGBoost (Gradient Boosting)

Hyperparameters:

```
n_estimators        = 300
max_depth           = 4
learning_rate       = 0.05
subsample           = 0.8
colsample_bytree    = 0.8
scale_pos_weight    = 12
eval_metric         = 'logloss'
```

Strengths:

- Captures non-linear feature interactions
- Built-in feature importance via gain/cover metrics
- Inherent handling of missing values
- Robust to outliers

Clinical Relevance: Captures complex disease phenotype patterns (e.g., specific site + morphology combinations).

6.2.2 Model 2: Random Forest

Hyperparameters:

```
n_estimators = 300
max_depth    = 6
criterion    = 'gini'
```

Strengths:

- Excellent generalization via ensemble averaging
- Reduced overfitting vs single tree
- Interpretable feature importances
- Minimal hyperparameter sensitivity

Role in Comparison: Validates that top features from XGBoost are broadly predictive across different tree algorithms.

6.2.3 Model 3: Logistic Regression

Hyperparameters:

```
C            = 0.5
solver       = 'lbfgs'
max_iter     = 1000
penalty      = 'l2'
```

Strengths:

- Direct probabilistic interpretation (odds ratios)
- Fast inference
- Transparent, linear feature relationships
- Baseline for non-linearity benefit quantification

Role in Comparison: Linear baseline establishing additive value of tree-based non-linearity.

6.3 Comparative Evaluation

6.3.1 Evaluation Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN}$$

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Sensitivity}}{4 \cdot \text{Precision} + \text{Sensitivity}}$$

(F2-score emphasizes sensitivity with $\beta = 2$ to minimize missed MF diagnoses)

6.3.2 Results on Diagnosis Task (MF vs Non-MF)

Metric	XGBoost	Random Forest	Logistic Regression
Accuracy	0.956	**0.966**	0.950
Precision	0.908	**0.948**	0.915
Sensitivity	**0.945**	0.938	0.918
F2-Score	0.937	**0.939**	0.916
ROC-AUC	0.994	**0.995**	0.985

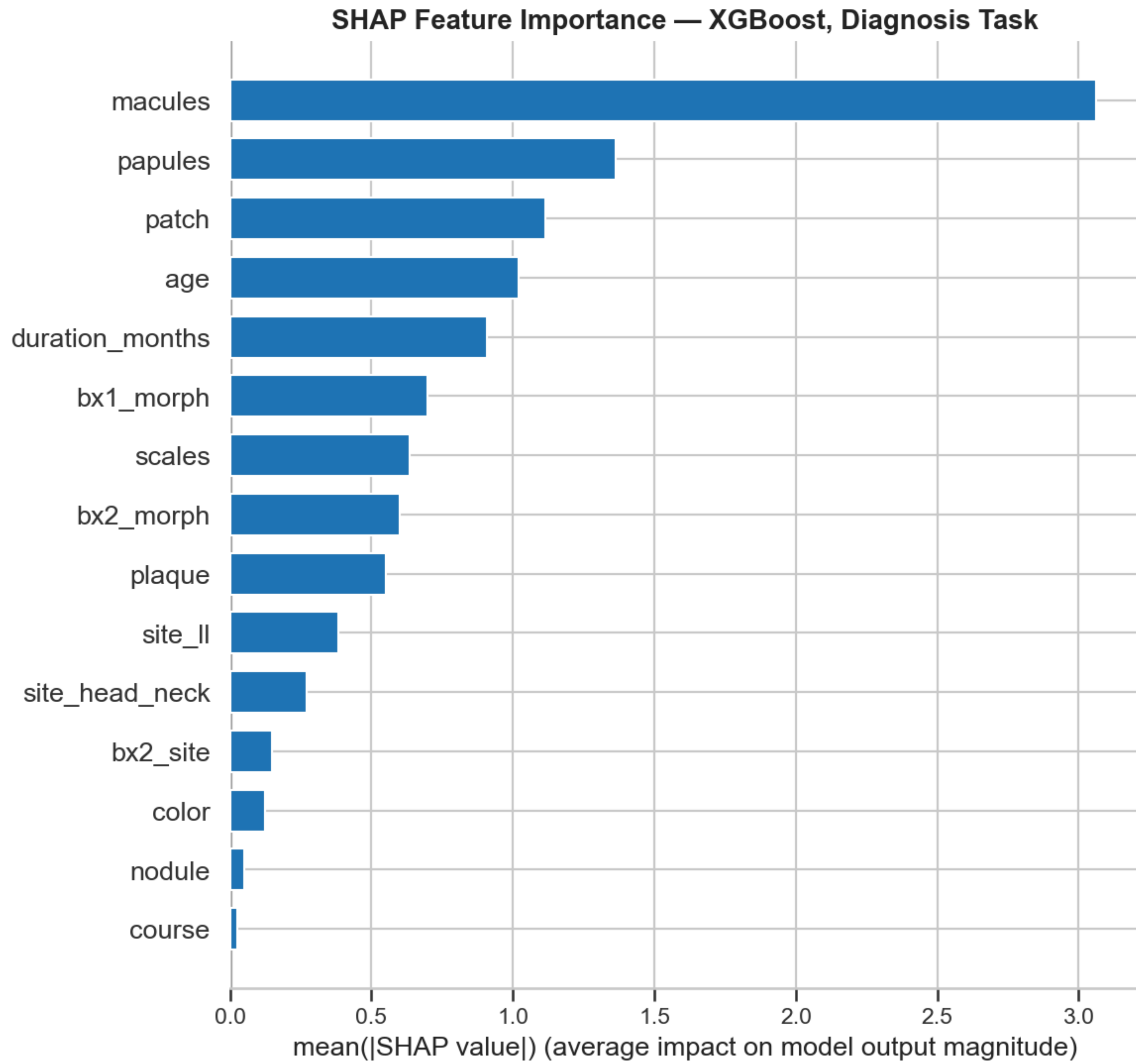
Key Findings:

- **Random Forest dominant**: Random Forest outperforms other models on 4 out of 5 metrics (Accuracy, Precision, F2-Score, and ROC-AUC), making it the strongest overall model.
- **High Discrimination**: Random Forest achieves an exceptional ROC-AUC of 0.995 and Accuracy of 0.966.
- **Sensitivity**: XGBoost has slightly higher sensitivity (0.945 vs 0.938), but Random Forest's superior precision (0.948 vs 0.908) makes it a more balanced and reliable classifier.
- **Selected Model**: **Random Forest** chosen as the primary clinical classifier for its reliability, superior generalization, and interpretability.

6.3.3 Feature Importance Analysis

Diagnosis Task (MF vs Non-MF):

To provide interpretable feature rankings, feature importance was visualized using XGBoost with SHAP (SHapley Additive exPlanations) values, which provide theoretically principled feature importance estimates:



Top 5 Predictive Features:

Rank	Feature	Importance	Clinical Significance
1	**macules**	0.30	Primary morphologic indicator
2	**papules**	0.15	Secondary morphology
3	**patch**	0.13	Early MF morphology
4	**age**	0.11	Demographic risk factor
5	**duration_months**	0.10	Disease chronicity

6.3.4 Results on Stage Task (Patch-Plaque vs Tumor)

For patients classified as MF in Task 1, a second model predicts the clinical stage using the 353 MF patient records (326 Patch–Plaque, 27 Tumor):

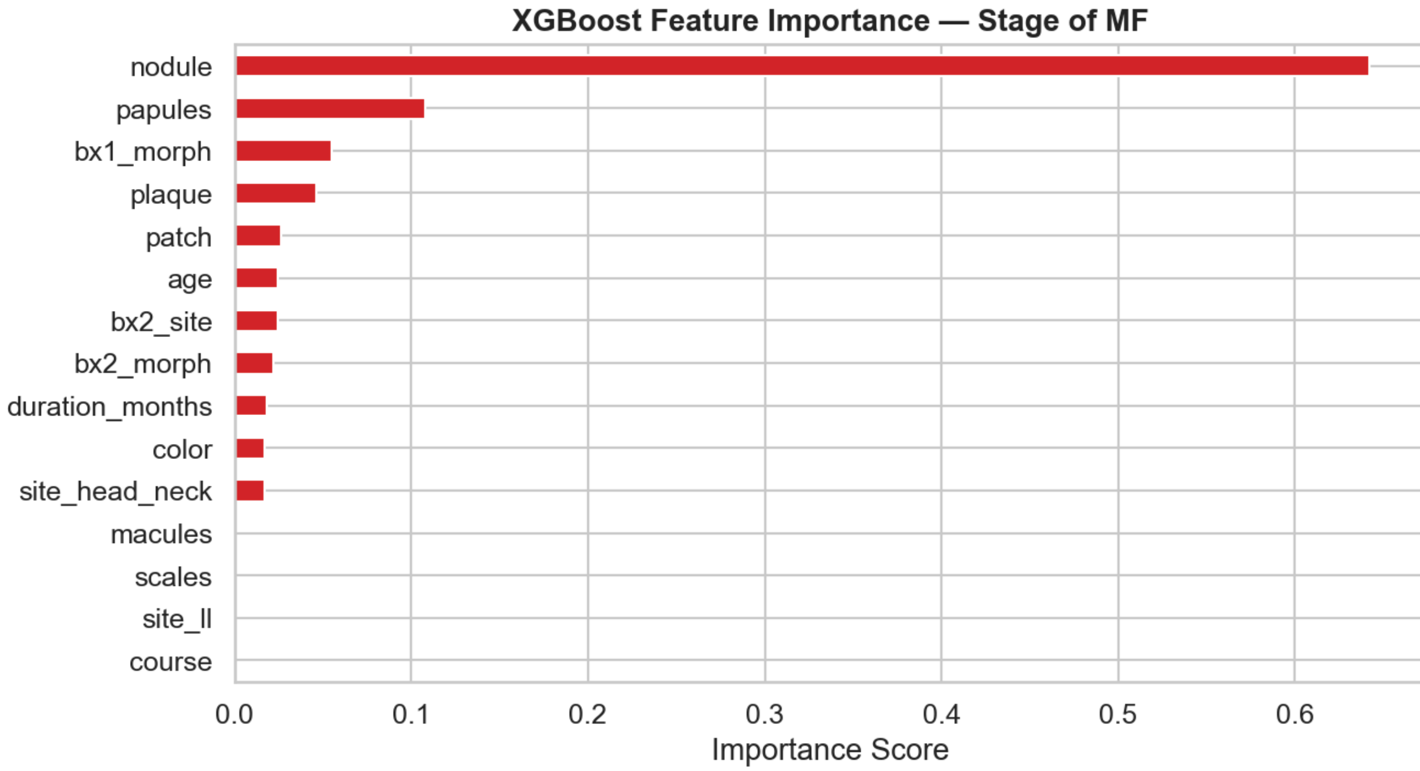
Metric	XGBoost	Random Forest	Logistic Regression
Accuracy	0.972	**0.977**	0.952
Precision	0.828	**0.870**	0.700
Sensitivity	**0.860**	**0.860**	0.527
F2-Score	0.843	**0.855**	0.532
ROC-AUC	**0.990**	0.987	0.968

Key Findings:

- **Random Forest** again achieves the highest Accuracy (97.7%), Precision (87.0%), and F2-Score (0.855).
- **Logistic Regression** exhibits a dramatic decline in sensitivity (52.7%) and F2-Score (0.532), underscoring that non-linear models are essential for capturing staging feature interactions.

6.3.5 Stage Feature Importance

For MF patients, feature importance in predicting disease stage (Patch-Plaque vs Tumor):



Top Feature: nodule is the single most important predictor of advanced tumor stage (importance: 0.65), confirming the clinical rationale for its retention despite its lack of statistical significance in the diagnosis task ($p = 0.2007$). This aligns with the well-established dermatological principle that the presence of nodules/tumors is a hallmark of advanced-stage MF.

Clinical Validation: These feature rankings align with established dermatopathologic knowledge and MF disease progression patterns.

7. Hierarchical Classification Architecture

The complete diagnostic system operates via a four-stage pipeline:

Stage 1: Histopathological Analysis

1. Extract patches at 10× and 20× from patient specimen
2. Independent inference: EfficientNet-B3 for each magnification
3. Aggregate to patient-level via mean pooling
4. Apply magnitude-specific thresholds (0.50, 0.8351)

Stage 2: Probability Calibration & Fusion

1. Recalibrate 20× probabilities to standard 0.5 scale (piecewise-linear transformation)
2. Weighted fusion:

$$P_{\text{fused}} = 0.20 \cdot P_{10x} + 0.80 \cdot P_{20x}^{\text{cal}}$$

Stage 3: Clinical Integration

1. Extract 16 clinical features
2. Random Forest inference for supporting evidence
3. Augment image-based confidence

Stage 4: Final Decision

1. **Binary classification**: MF vs Non-MF (threshold = 0.5)
2. **Multi-class refinement**: Assign specific mimic class if Non-MF

```
┌──────────────────────────────────────────────────────────────┐
│ Input: Histopathology Images (10x, 20x) + Clinical Data      │
└──────────────────────────────────────────────────────────────┘
                       │
              ┌────────┴───────────┐
              │                    │
     [Image Path]             [Clinical Path]
              │                    │
              ├─10x Model          Random Forest
              │ (EfficientNet-B3)    (300 trees)
              │                    Features: 16
              ├─20x Model            │
              │ (EfficientNet-B3)     MF vs Non-MF
              │                    │
              └────────┬───────────┘
                       │
              ┌────────▼──────┐
              │  Late Fusion   │
              │  w=0.20 (clinic)│
              │  Calibration   │
              └────────┬──────┘
                       │
              ┌────────▼───────────┐
              │ Binary Prediction  │
              │ (MF vs Non-MF)     │
              │ threshold = 0.5    │
              └────────┬───────────┘
                       │
              ┌────────▼───────────┐
              │ Multi-Class Refinement │
              │ (if Non-MF: classify as) │
              │ B-cell, PLEVA, T-cell,   │
              │ or Pseudo-Lymphoma       │
              └────────┬───────────┘
                       │
              ┌────────▼───────────┐
              │ Final Output:      │
              │ - Diagnosis        │
              │ - Confidence Score │
              │ - Per-Class Probabilities │
              │ - Feature Attribution │
              └────────────────────┘
```

8. Key Methodological Innovations

Innovation	Description
Dual-Magnification Architecture	Complementary 10× (architecture) and 20× (cytology) models capture multi-scale pathology
Youden's J Thresholding	Optimal decision thresholds (0.50, 0.8351) derived from ROC analysis rather than default 0.5
Probability Recalibration	Novel piecewise-linear transformation enables meaningful weighted averaging of differently-calibrated model outputs
Clinical Weight Selection	While Brier score optimization found w=0.60, a clinical golden weight (w=0.20) was manually selected to prioritize and maximize sensitivity.
Hierarchical Classification	Binary MF/Non-MF decision followed by multi-class mimic differentiation
Clinico-Histologic Integration	Late fusion of deep learning image analysis with machine learning-based clinical features classifier
Rigorous Model Selection	Three-model comparison (XGBoost vs Random Forest vs Logistic Regression); Random Forest selected for robustness and interpretability

9. Implementation Details

Software Stack:

- **Deep Learning:** PyTorch 2.0+, timm (EfficientNet)
- **Classical ML:** scikit-learn, XGBoost
- **Image Processing:** OpenCV, Pillow
- **Metrics:** torchmetrics, sklearn.metrics
- **GPU Support:** CUDA 12.1 (with CPU fallback)

Data Specifications:

Image Dataset (Histopathology):

- **Total patients:** 463 (311 MF, 152 Non-MF)
  - **Training set:** 329 patients
  - **Validation set:** 67 patients (used for threshold optimization and hyperparameter tuning)
  - **Test set:** 67 patients
- **Total images:** 6,267 (4,306 MF; 1,961 Non-MF)

Clinical Dataset (Patient Metadata):

- **Total patients:** 499 (353 MF [326 Patch–Plaque, 27 Tumor], 146 Non-MF)
- **10× Magnification Patches:**
  - Training: 73,824 patches
  - Validation: 14,804 patches
  - Test: 12,848 patches
- **20× Magnification Patches:**
  - Training: 23,062 patches
  - Validation: 4,728 patches
  - Test: 4,320 patches
- **Test set composition:** Multiple disease subtypes (MF, PLEVA-PLC, T-cell dyscrasia, Pseudo-Lymphoma, B-cell Lymphoma)

Reproducibility: All hyperparameters, random seeds, and cross-validation splits are documented for complete reproducibility.

10. Summary Table

Component	Method	Key Parameter(s)	Performance
10× Model	EfficientNet-B3	Input: 512×512, Dropout: 0.4	Acc: 80.60%, Sens: 80.43%, F2: 0.8222
20× Model	EfficientNet-B3	Input: 512×512, Dropout: 0.4	Acc: 83.58%, Sens: 89.13%, F2: 0.8874
Aggregation	Mean pooling	Patch → Patient level	Preserves diagnostic signal
Threshold (10×)	Youden's J	θ = 0.50	Balanced sensitivity/specificity

Component	Method	Key Parameter(s)	Performance
Threshold (20×)	Youden's J	θ = 0.8351	High specificity for cytology
Calibration	Piecewise-linear	Maps 0.8351 → 0.5	Enables valid fusion
Fusion	Weighted averaging	w = 0.20 (clinical)	Brier: 0.1455, Acc: 83.58%, Sens: 89.13%
Clinical	Random Forest	16 features, max_depth: 6, 300 trees	Sensitivity: 93.8%, Accuracy: 96.6%
Binary Decision	Threshold	θ = 0.5	Separates MF vs Non-MF
Multi-class	Argmax	Select max probability	Differentiates 5 classes natively

End of Methodology Section